

QUESTION BANK
CHAPTER 8-ELECTROMAGNETIC WAVES
SECTION A

1 MARK QUESTIONS

- Q1. What is displacement current?
- Q2. What are electromagnetic waves?
- Q3. State the Maxwell's law of induction.
- Q4. Which of the following is false for electromagnetic waves:
- (a) transverse (b) non-mechanical waves
(c) longitudinal (d) produced by accelerating charges
- Q5. If the magnetic monopole exists, then which of the Maxwell's equation to be modified?
- (a) $\oint \mathbf{E} \cdot d\mathbf{S} = q / \epsilon_0$ (b) $\oint \mathbf{B} \cdot d\mathbf{S} = 0$
(c) $\oint \mathbf{E} \cdot d\mathbf{l} = - d\phi_B / dt$ (d) $\oint \mathbf{E} \cdot d\mathbf{l} = \mu_0(I_c + I_d)$
- Q6. Which of the following is NOT true for electromagnetic waves?
- (a) They transport energy
(b) They have momentum
(c) They travel at different speeds in air depending on their frequency
(d) They travel at different speeds in medium depending on their frequency
- Q7. Why can light travel in vacuum, whereas sound cannot do so?
- Q8. Is displacement current, like conduction current, a source of magnetic field?
- Q9. Expand the acronym LASER and RADAR.
- Q10. Which of the following statement is **NOT true** about the properties of electromagnetic waves?
- (a) These waves do not require any material medium for their propagation
(b) Both electric and magnetic field vectors attain the maxima and minima at the same time
(c) The energy in electromagnetic wave is divided equally between electric and magnetic fields
(d) Both electric and magnetic field vectors are parallel to each other.
- Q11. Do Electromagnetic waves carry energy and momentum? **(CBSE2017)**
- Q12. Identify the EM waves whose wavelength vary as
- (a) $10^{-12}\text{m} < \lambda < 10^{-8}\text{m}$

(b) $10^{-3}\text{m} < \lambda < 10^{-1}\text{m}$

Write one use for each.

(CBSE 2017)

Q13. Name the Electromagnetic radiations used for

(i) Water purification (ii) Eye surgery

(CBSE 2018)

Q14. How is displacement current produced between plates of parallel plates during charging?

(CBSE 2020)

ASSERTION REASON TYPE QUESTIONS

Directions: In each of the following questions, a statement of Assertion (A) is given followed by a corresponding statement of Reason (R) just below it. Of the statements, mark the correct answer as:

(A) If both assertion and reason are true and reason is the correct explanation of assertion.

(B) If both assertion and reason are true but reason is not the correct explanation of assertion.

(C) If the assertion is true and the reason is false.

(D) If both assertion and reason are false.

Q15. **Assertion:** Electromagnetic waves are transverse in nature.

Reason: The electric and magnetic fields are perpendicular to each other and perpendicular to the direction of propagation.

Q16. **Assertion:** The electromagnetic wave is transverse in nature.

Reason: Electromagnetic waves propagate parallel to the direction of electric and magnetic fields.

Q17. **Assertion:** The velocity of electromagnetic waves depends on electric and magnetic properties of the medium.

Reason: Velocity of electromagnetic waves in free space is constant.

CASE STUDY BASED QUESTIONS

Q18. LASER

Electromagnetic radiation is a natural phenomenon found in almost all areas of daily life, from radio waves to sunlight to x-rays. Laser radiation – like all light – is also a form of electromagnetic radiation. Electromagnetic radiation that has a wavelength between 380 nm and 780 nm is visible to the human eye and is commonly referred to as light. At wavelengths longer than 780 nm, optical radiation is termed infrared (IR) and is invisible to the eye. At wavelengths shorter than 380 nm,

optical radiation is termed ultraviolet (UV) and is also invisible to the eye. The term “laser light” refers to a much broader range of the electromagnetic spectrum than just the visible spectrum, anything between 150 nm up to 11000 nm (i.e. from the UV up to the far IR). The term laser is an acronym which stands for “light amplification by stimulated emission of radiation”.

Einstein explained the stimulated emission. In an atom, an electron may move to a higher energy level by absorbing a photon. When the electron comes back to the lower energy level it releases the same photon. This is called spontaneous emission. This may also so happen that the excited electron absorbs another photon, releases two photons and returns to the lower energy state. This is known as stimulated emission.

Laser emission is therefore a light emission whose energy is used, in lithotripsy, for targeting and ablating the stone inside the human body organ.

Apart from medical usage, lasers are used for optical disk drives, printers, barcode readers etc.

(I) What is the full form of LASER?

- (a) Light amplified by stimulated emission of radiation
- (b) Light amplification by stimulated emission of radiation
- (c) Light amplification by simultaneous emission of radiation
- (d) Light amplified by synchronous emission of radiation

(II) The “stimulated emission” is the process of:

- (a) release of a photon when an electron comes back from higher to lower energy level.
- (b) release of two photons by absorbing one photon when the electron comes back from higher to lower energy level.
- (c) absorption of a photon when an electron moves from lower to higher energy level.
- (d) None of the above

(III) What is the range of amplitude of LASER?

- (a) 150 nm – 400 nm (b) 700 nm – 11000 nm (c) Both the above (d) None of the above

(IV) Lithotripsy is:

- (a) an industrial application. (b) a medical application.
- (c) laboratory application. (d) process control application.

(V) LASER is used in:

- (a) optical disk drive. (b) transmitting satellite signal.
- (c) radio communication. (d) ionization.

2 MARKS QUESTIONS

Q19. Suggest reasons, why

- (a) food in metal containers cannot be cooked in a microwave oven.
- (b) an empty glass container does not get hot in a microwave oven.

Q20. Electromagnetic waves with wavelength

- (i) λ_1 is suitable for radar systems used in aircraft navigation.
- (ii) λ_2 is used to kill germs in water purifiers.
- (iii) λ_3 is used to improve visibility in runways during fog and mist conditions.

Identify and name the part of the electromagnetic spectrum to which these radiations belong. Also arrange these wavelengths in ascending order of their magnitude.

3 MARKS QUESTIONS

Q21. Give two uses of 1) IR radiation 2) Microwaves 3) UV radiation

Q22. Write down the properties of electromagnetic waves.

Q23. Write short notes on 1) Microwave 2) X-ray 3) Radio waves

SECTION A-ANSWER KEY

1 MARK QUESTIONS

1.The current which comes into play in the region in which the electric field (or electric flux) is changing with time is called displacement current.

$$I_d = \epsilon_0 d\phi_E / dt$$

2.The accelerated charge produces the electromagnetic waves. They have the transverse nature. The electromagnetic waves are a non-mechanical wave which moves with speed equals to the speed of light.

3. Whenever changing the electric flux (Φ_E) in the region encircled by loop, then magnetic field is induced.

4(c)

5. (b)

6 (c)

7.Light waves are electromagnetic in nature. They do not require a material medium for propagation. So, the light can travel in vacuum. On the other hand, sound waves require a material medium for propagation. They are mechanical waves and cannot travel in vacuum.

8. Yes, displacement current is also a source of magnetic field like the conduction current .

9. LASER-Light amplification by stimulated emission of radiation

RADAR-Radio detection and ranging.

10. (iv) Both electric and magnetic field vectors are parallel to each other.

11. Yes, EM waves carry both energy and momentum.

12. (a)X rays - used in in detection of fractures in bones.

(b)Gamma rays- used in radio therapy for treatment of cancer.

13. UV rays

14. Refer Ncert

ASSERTION REASON BASED QUESTIONS

15. (a)

16. (c)

17.(b)

CASE STUDY BASED QUESTIONS

18. (I)B (II)B (III) C (IV) B (V) A

2 MARKS QUESTIONS

19. In a microwave oven, the frequency of microwaves is selected to match the resonance frequency of water molecules, so that the energy from the waves is transferred efficiently to the kinetic energy of the molecules. This raises the temperature of any food containing water. (a) The atoms of the metallic container are set into forced vibrations by the microwaves. Due to this, energy of the microwaves is not efficiently transferred to the metallic container. Owing to this, food in metallic containers cannot be cooked in a microwave oven.

(b) The molecules of the glass container do not respond to the frequency of microwaves. Due to this, energy from the microwaves is not transferred to the glass container and hence it does not get hot in a microwave oven.

20. λ_1 -Microwave

λ_2 - ultraviolet

λ_3 - infrared

Ascending order - $\lambda_2 < \lambda_3 < \lambda_1$

3 MARKS QUESTIONS

21. IR radiation • It is used in infrared photography • TV remote as a signal carrier • Heat therapy for muscular pain or sprain.

Microwaves • It is used in Radio and Television communication system • It is used in cellular phones (Voice communication)

UV radiation • It is used to destroy bacteria in sterilizing the surgical instruments. • It is used in Burglar alarm • It is used to detect the invisible writing, finger prints.

22. Refer notes of study material/Ncert

23.(i) Microwaves are suitable for RADAR systems that are used in aircraft navigation. These rays are produced by special vacuum tubes, namely klystrons and magnetron diodes. (ii) Infrared rays are used to treat muscular strain. These rays are produced by hot bodies and molecules.

(iii) X-rays are used as a diagnostic tool in medicine. These rays are produced, when high energy electrons are stopped suddenly on a metal of high atomic number.

SECTION B

1 MARK QUESTIONS

Q1.If the amplitude of the magnetic field is 3×10^{-6} tesla, then the amplitude of the electric field for a electromagnetic waves is

- (a) 100 V/m (b) 300 V/m (c) 600 V/m (d) 900 V/m

Q2.In an electromagnetic wave travelling in free space the rms value of the electric field is 3 V/m. The peak value of the magnetic field is,

- (a) 1.414×10^{-8} T (b) 1.0×10^{-8} T (c) 2.828×10^{-8} T (d) 2.0×10^{-8} T

Q3.The direction of propagation of electromagnetic waves is along

- (a) $\mathbf{E}$ (b) $\mathbf{B}$ (c) $\mathbf{E.B}$ (d) $\mathbf{E \times B}$

Q4.In an electromagnetic wave:

- (a) power is transmitted along the magnetic field
(b) power is transmitted along the electric field
(c) power is equally transferred along the electric and magnetic field
(d) power is transmitted in a direction perpendicular to both the fields.

Q5.Why are infrared waves called heat waves. Explain. What do you understand by the statement “Electromagnetic waves transport momentum”?
(CBSE 2018)

ASSERTION REASON TYPE QUESTIONS

Q6.**Assertion:** Infrared radiation plays an important role in maintaining the average temperature of earth.

Reason: Infrared radiations are sometimes referred to as heat waves

Q7. **Assertion:** When a charged particle moves in a circular path, it produces electromagnetic waves.

Reason: Charged particle has acceleration.

Q8. **Assertion:** Sound waves cannot travel in vacuum but light can travel in vacuum

Reason : Sound waves are longitudinal waves and they cannot be polarised but electromagnetic waves are transverse and they can be polarised.

CASE STUDY BASED QUESTIONS

Q9. The electromagnetic spectrum consists of visible light, x-rays, gamma rays, microwaves, ultraviolet rays, radio waves and infrared waves. The waves used in radio and television communication are the radio waves having frequency range 500 kHz to 1000MHz. In the ultrahigh frequency bands cellular phones use the radio waves to transmit the voice. Microwaves are the waves having short wavelengths. In aircraft navigation, for the radar system microwaves are used due to their short wavelength. Infrared waves are also called heat waves. Infrared radiation has the most importance in maintaining earth's surface temperature through the greenhouse effect. The infrared waves have vast applications in real life such as infrared detectors that are used for military purposes and also to see the growth of crops. The waves which are visible to the human eye are the visible rays. Visible rays are having frequency range as 4×10^{14} Hz to 7×10^{14} Hz. The huge source of ultraviolet light is the sun. Ultraviolet rays have wavelength range from 4×10^{-7} m to 6×10^{-10} m. X-rays are the rays having most importance in medical applications which have a wavelength range 10nm to 10^{-4} nm. X-rays are used to destroy the living tissue and organisms in the medical field. Then gamma rays are the rays having wavelength range as 10^{-10} m to 10^{-14} m which are the high frequency radiations mostly produced in nuclear reactions. Gamma rays are also used to destroy cancer cells in the medical field.

(I) The TV waves range from _____ which are the radio waves.

- (a) 54Hz to 890Hz (b) 54 MHz to 890 MHz
(c) 500 kHz to 1000 MHz (d) 1000 Hz to 1000 KHz

(II) The domestic application of microwaves used is _____

- (a) electric induction (b) water heater (c) TV (d) microwave oven

(III) The part of the electromagnetic spectrum which is detected by human eye is having wavelength range as

- a) 900 – 400 nm b) 200 – 400 mm c) 700 – 400 mm d) 700 – 400 nm

(IV) Why are infrared waves also called heat waves?

(V) How does the ozone layer in the atmosphere play a protective role?

2 MARKS QUESTIONS

Q10. Why are infrared radiations also referred as heat waves? Write the name of radiations which lie next to infrared radiations in the electromagnetic spectrum. (CBSE2020)

Q11. A plane electromagnetic wave travels in vacuum along z-direction. What can you say about the directions of its electric and magnetic field vectors? If the frequency of the wave is 30 MHz, what is its wavelength?

Q12. (a) Identify the part of EM spectrum used in (i) RADAR (ii) Eye surgery. Write the frequency range. **(CBSE 2019)**

(b) Prove that the average energy density of oscillating Electric field is equal to that of oscillating magnetic field?

3 MARKS QUESTIONS

Q13. Answer the following questions.

- (i) Show, by giving a simple example, how EM waves carry energy and momentum.
- (ii) How are microwaves produced? Why is it necessary in microwaves ovens to select the frequency of microwaves to match the resonant frequency of water molecules?
- (iii) Write two important uses of infrared waves.

Q14. Name the parts of the electromagnetic spectrum which is

- (i) suitable for RADAR systems in aircraft navigations.
- (ii) used to treat muscular strain.
- (iii) used as a diagnostic tool in medicine. Write in brief, how these waves can be produced?

Q15. (i) Arrange the following electromagnetic waves in the descending order of their wavelengths.

(a) Microwaves (b) Infrared rays (c) Ultraviolet radiation (d) g-rays

(ii) Write one use each of any two of them

Q16. Explain the Maxwell's modification of Ampere's circuital law.

Q17. Discuss the source and propagation of electromagnetic waves.

ANSWERS (SECTION B)

1 MARK QUESTIONS

1. (d) 2. (a) 3. (d) 4. (d)

5. Infrared waves are known as heat waves because they produce heat.

When Electromagnetic waves hit body the mass is lost by the momentum is conserved that is transferred from Electromagnetic waves to the body.

ASSERTION REASON BASED QUESTIONS

6.B 7.B 8.B

CASE STUDY BASED QUESTIONS

9.(I)(b) 54MHz to 890 MHz

(II)(d) microwave oven

(III)(d) 700 – 400 nm

(IV) refer study material/Ncert

(V) The very harmful ultraviolet rays coming from the sun are absorbed by the ozone layer in the atmosphere which is at an altitude of 40 – 50 km from the earth's surface. And due to which very less ultraviolet light rays from the sun reach the earth. Ultraviolet rays are very harmful to humans which may cause skin cancer. In this way, the ozone layer in the atmosphere plays a very protective role.

2 MARKS QUESTIONS

10. Infrared red waves are the waves which are having frequency lower than the frequency of visible light range. This infrared wave causes the electrons, whole atoms or molecules also of that material. Due to such increased vibrations the internal energy of the material gets increased and thereby increases in the temperature of the material and heat is generated. Because of this reason infrared waves are also called the heat waves.

Visible rays come next to infrared radiation in electromagnetic spectrum having shorter wavelength.

The radiations next to infrared radiations, having longer wavelengths will be microwaves

11. As we know that, the direction of electromagnetic wave is perpendicular to both electric and magnetic fields. Here, electromagnetic wave is travelling in z-direction, then electric and magnetic fields are in xy-direction and are perpendicular to each other. Frequency of waves, $n = 30 \text{ MHz} = 30 \times 10^6 \text{ Hz}$ Speed, $c = 3 \times 10^8 \text{ m/s}$ Using the formula, $c = n \cdot \lambda$ Wavelength of electromagnetic

waves, $\lambda = c/n = (3 \times 10^8)/(30 \times 10^6) = 10 \text{ m}$ Thus, the wavelength of electromagnetic waves is 10 m.

12.(a)(i) Microwaves range-1GHz-2GHz (ii) UV rays-range 10^{15} - 10^{17} Hz

$$(b) U_E = \frac{1}{2} \epsilon_0 E^2 \quad U_B = \frac{B^2}{2} \quad ; c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

$$E = BC ; \text{ Thus } U_E = \frac{1}{2} \epsilon_0 (BC)^2 = \frac{B^2}{2 \mu_0}$$

3 MARKS QUESTIONS

13.(i) Consider a plane perpendicular to the direction of propagation of the wave. An electric charge, on the plane will be set in motion by the electric and magnetic fields of EM wave, incident on this plane. This is only possible, if EM wave constitutes momentum and energy. Thus, this illustrates that EM waves carry energy and momentum.

(ii) Microwaves are produced by special vacuum tube like the klystron, magnetron and Gunn diode. The frequency of microwaves is selected to match the resonant frequency of water molecules, so that energy is transformed efficiently to increase the kinetic energy of the molecules. Thus, facilitating the food to cook properly.

(iii) Uses of infrared rays (a) In knowing the molecular structure and therapy to heal muscular pain. (b) In remote control of TV, VCR, etc.

14. **Microwave** It is produced by special vacuum tubes such as klystron, magnetron and gun diode. The frequency range of microwaves is 10^9 Hz to 10^{11} Hz. These waves undergoes reflection can be polarized. It is used in radar system for aircraft navigation, speed of the vehicle, microwave oven for cooking and very long distance wireless communication through satellite.

☞ **X-Ray** It is produced when there is sudden stopping of high-speed electrons at high atomic number target, and also by electronic transitions among the innermost orbits of atoms. The frequency range of X-rays is from 10^{17} Hz to 10^{19} Hz. X-rays have more penetrating power than ultraviolet radiation. X-rays are used extensively in studying structures of inner atomic electron shells and crystal structures. It is used in detecting fractures, diseased organs, formation of bones and stones, observing the progress of healing bones. Further, in a finished metal product, it is used to detect faults, cracks, flaws and holes.

☞ **Radio Waves** They are produced by accelerated motion of charges in conducting wires. The frequency range is from a few Hz to 10^9 Hz. They show reflection and diffraction

15.(i) The decreasing order of wavelengths of electromagnetic waves is Microwaves > Infrared > Ultraviolet radiation > γ -rays (ii) Microwaves -They are used in RADAR devices. γ -rays- It is used in radio therapy.

16.Refer study material

17.Refer study material

SECTION C

1 MARK QUESTIONS

Q1. The relative magnetic permeability of the medium is 2.5 and the relative electrical permittivity of the medium is 2.25. Compute the refractive index of the medium.

- (a) 2.37 (b) 4.75 (c) 0.25 (d) 1.05

Q2. Compute the speed of the electromagnetic wave in a medium if the amplitude of electric and magnetic fields are 3×10^4 and 2×10^{-4} tesla, respectively

- (a) 3×10^8 m/s (b) 1.5×10^8 m/s (c) 6×10^8 m/s (d) 5×10^8 m/s

Q3. The charging current for a capacitor is 0.25 A. What is the displacement current across its plates?

Q4. If the Earth did not have an atmosphere, would its average surface temperature be higher or lower than what it is now?

ASSERTION REASON TYPE QUESTIONS

Q5. **Assertion:** The frequencies of incident, reflected and refracted beams of monochromatic light incident from one medium to another are the same.

Reason: The incident, reflected and refracted rays are coplanar.

Q6. **Assertion:** The earth without its atmosphere will be inhospitably cold.

Reason: All heat would escape in the absence of atmosphere.

Q7. **Assertion:** Microwaves are better carriers of signal than optical waves.

Reason: Microwaves move faster than optical waves.

Q8. **Assertion:** Gamma rays are more energetic than x rays.

Reason: Gamma rays are of nuclear origin but x rays are produced due to sudden deceleration of high energy electrons while falling in the metal of high atomic number.

CASE STUDY BASED QUESTION

Q9. According to Maxwell's electromagnetic equations it has been proved that electric and magnetic field vectors are perpendicular to each other and also perpendicular to the direction of propagation as shown in the figure below. If E_x is the electric field along X axis, then B_y will be the direction of magnetic field along y axis and both which are perpendicular to the z axis showing direction of propagation. The light waves are also electromagnetic waves and may travel through

vacuum also. So, we can find the velocity of a light traveling through the material medium having permittivity ϵ and magnetic permeability μ as $V = 1/\sqrt{\epsilon\mu}$

In this way, we proved that velocity of light also depends on the electrical and magnetic properties of that medium through which it is traveling. The velocity of light which is constant everywhere is having value as $3 \times 10^8 \text{ m/s}$. The most technological importance of electromagnetic waves is that they have a strong capacity to take energy from one place to another place. The best examples are radio waves, TV signals which also carry energy from their broadcasting stations. Also, life is possible on the earth only because of the sunlight coming from the sun to the earth which also carries energy and it is nothing but the electromagnetic waves. Due to which electromagnetic waves are considered as the transverse waves.

(i) The ratio of relative permittivity of the medium to the permittivity of vacuum is called as _____.

- (a) permeability (b) permittivity of free space
(c) dielectric constant of the medium (d) electric intensity

(II) Who showed that electromagnetic waves can be polarised?

- (a) Maxwell (b) Hertz (c) Ampere (d) Michelson and Morley

(III) The pressure exerted by the electromagnetic waves is called as

- (a) light pressure (b) electric pressure (c) magnetic pressure (d) radiation pressure

(IV) What is the relationship between magnitude of magnetic field and electric field in case of electromagnetic waves from Maxwell's equations?

(V) What is meant by the permittivity and permeability of the medium?

2 MARKS QUESTIONS

Q10. The electric field of an electromagnetic wave is given by $E = 50 \sin \omega(t - x/c)$ N/C. Find the energy contained in a cylinder of cross-section 10 cm^2 and length 50 cm along the X-axis.

Q11. A variable-frequency a.c. source is connected to a capacitor. Will the displacement current increase or decrease with increase in frequency?

3 MARKS QUESTIONS

Q12. The magnetic field in a plane electromagnetic wave is given by $B_y = 2 \times 10^{-7} \sin [0.5 \times 10^3 x + 1.5 \times 10^{11} t]$ (in T)

- (a) What is the wavelength and frequency of the wave?
(b) Write an expression for the electric field.

Q13. Answer the following questions-

- (i) Name the EM waves which are used for the treatment of certain forms of cancer. Write their frequency range.
- (ii) Thin ozone layer on top of stratosphere is crucial for human survival. Why?
- (iii) Why is the amount of the momentum transferred by the EM waves incident on the surface so small?

ANSWERS (SECTION C)

1 MARK QUESTIONS

1(a)

2(b)

3. Displacement current is always equal to conduction current and hence equals to 0.25 A

4. The infrared radiation emitted by earth is retained by the earth's atmosphere due to the greenhouse effect and this keeps the earth warm. If the Earth did not have an atmosphere, its average temperature would have been low.

ASSERTION REASON TYPE QUESTIONS

5. B

6. A

7. D

8. B

CASE STUDY BASED QUESTIONS

9. (I) (c) dielectric constant of the medium

(I) (b) Hertz

(II) (d) radiation pressure

(III) From Maxwell's equations, the relationship between magnitude of magnetic field and electric field is given as $B_0 = (E_0/c)$

(IV) Permittivity of the medium is nothing but the ability of that medium to store electric potential energy in that medium.

While permeability of the medium is also the ability of the medium to allow the number of magnetic field lines through it.

2 MARKS QUESTIONS

10. The average value of energy density (energy / volume) is given by

$U_{av} = \frac{1}{2} \epsilon_0 E_0^2$ / Total volume of the cylinder $V = A l$;

Total energy contained in the cylinder, $U = (U_{av}) / V = (\frac{1}{2} \epsilon_0 E_0^2) / A l$

$= \frac{1}{2} [(8.86 \times 10^{-12})(50)^2] / [(10 \times 10^{-4})(50 \times 10^{-2})] = 5.5 \times 10^{-12} \text{ J}$

11. If we increase the frequency, the reactance of the capacitor will decrease and consequently conduction current will increase. Since displacement current is equal to conduction current, the displacement current will increase with increase in frequency of a.c. source.

3 MARKS QUESTIONS

12. Here, $B_y = 2 \times 10^{-7} \sin [0.5 \times 10^3 x + 1.5 \times 10^{11} t]$

(a) The Y-component of the magnetic field is given by $B_y = B_0 \sin 2\pi \left(\frac{x}{\lambda} + \frac{t}{T} \right)$

Comparing the given equation with the above equation:

$$2\pi/\lambda = 1/(0.5 \times 10^3 \lambda) = 1.257 \times 10^{-2} \text{ m}$$

$$\text{Also } 2\pi/T = 1.5 \times 10^{11} \text{ Or } \nu = 2.387 \times 10^{10} \text{ Hz}$$

(b) Since the argument of sine in the expression for the magnetic field is of the type $(kx + \omega t)$, the direction of propagation of the e. m. wave is along negative X-axis and the magnetic field is along negative Y-axis.

Hence, the electric field is along negative Z-axis and expression for it is given by

$$E_y = E_0 \sin 2\pi \left(\frac{x}{\lambda} + \frac{t}{T} \right)$$

$$\text{Here, } E_0 = B_0 c = 2 \times 10^{-7} \times 3 \times 10^8 = 60 \text{ V/m;}$$

$$E_z = 60 \sin [0.5 \times 10^3 x + 1.5 \times 10^{11} t] \text{ (in V/m)}$$

13. i) γ -rays are used for the treatment of certain forms of cancer. Its frequency range is 3×10^{19} Hz to 5×10^{22} Hz.

(ii) The thin ozone layer on top of stratosphere absorbs most of the harmful ultraviolet rays coming from the sun towards the earth. They include UVA, UVB and UVC radiations, which can destroy the life system on the earth. Hence, this layer is crucial for human survival.

(iii) An electromagnetic wave transports linear momentum as it travels through space. If an electromagnetic wave transfers a total energy U to a totally absorbing surface in time t , then total linear momentum delivered to the surface. This means, the momentum range of EM waves is 10^{-19} to 10^{-41} . Thus, the amount of momentum transferred by the EM waves incident on the surface is very small.

-----*****-----